Title: Multi-scale method for ill-conditioned linear systems.

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1. Introduction:

Solving linear systems of equations is a fundamental problem in scientific computing, engineering & applied mathematics. The problem can be formally stated as finding a vector x that satisfies the linear equation in a matrix form:

$$Ax = b \tag{1}$$

where

$A = [a_{ij}] \in \mathbb{R}^{n \times n}$ is non singular,

$x = [x_j] \in \mathbb{R}^n$ unknown, &

$b \in \mathbb{R}^n$ is a constant vector.

There are 2 approaches for solving (1):

(i) direct methods (LU decomposition)

(ii) iterative methods (Jacobi, Gauss-Siedel, SOR, GMRES, ...)

For large sparse (contains a lot of zero entries) systems, iterative methods are popular & efficient.

However, it's difficult to develop effective methods for ill-conditioned linear systems whose coefficient matrices have large condition numbers:

$$\mathrm{cond}_{\boxed{\cdot}}(A) = \|A\|_{\boxed{\cdot}} \cdot \|A^{-1}\|_{\boxed{\cdot}}$$

(We usually consider 2-norm unless specified.)

If the cond(A) is so large, the solution of the system (1) is so sensitive to small changes in the input data. We call this type of sytems ill-conditioned & this project explores a way of improving condition number, cond(A).

2. Multi-scale method:

If the matrix A of equations (1) is ill-conditioned the system (1) can be solved by the following equations:

$$(2) \qquad \tilde{A}\,\tilde{x} = b \qquad \text{where}$$

$$\tilde{A} = \begin{bmatrix} \frac{1}{\|\alpha_1\|_2} a_{11} & \frac{1}{\|\alpha_2\|_2} a_{12} & \cdots & \frac{1}{\|\alpha_n\|_2} a_{1n} \\ \frac{1}{\|\alpha_1\|_2} a_{21} & \frac{1}{\|\alpha_2\|_2} a_{22} & \cdots & \frac{1}{\|\alpha_n\|_2} a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{\|\alpha_1\|_2} a_{n1} & \frac{1}{\|\alpha_2\|_2} a_{n2} & \cdots & \frac{1}{\|\alpha_n\|_2} a_{nn} \end{bmatrix}; \quad \tilde{x} = \begin{bmatrix} \|\alpha_1\|_2 x_1 \\ \|\alpha_2\|_2 x_2 \\ \vdots \\ \|\alpha_n\|_2 x_n \end{bmatrix}$$

Here α_j, $j = 1, 2, \ldots n$, is the jth column of A.

with defined (2), the following multi-scale method is obtained for sytem (1):

The Multi-scale method:

Step 1: Solve $\tilde{A}\tilde{x} = b$ with iterative methods.

Step 2:

$$(3) \qquad x = \begin{bmatrix} \frac{1}{\|\alpha_1\|_2}\tilde{x}_1 \\ \frac{1}{\|\alpha_2\|_2}\tilde{x}_2 \\ \vdots \\ \frac{1}{\|\alpha_n\|_2}\tilde{x}_n \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{\|\alpha_1\|_2} & 0 & \cdots & 0 \\ 0 & \frac{1}{\|\alpha_2\|_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \\ 0 & 0 & \cdots & \frac{1}{\|\alpha_n\|_2} \end{bmatrix}}_{D} \cdot \underbrace{\begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_n \end{bmatrix}}_{\tilde{x}}$$

By this multi-scale method, we could improve the condition number of the coefficient matrix A.

i.e. (to be proved later)

$$(4) \qquad \text{cond}(\tilde{A}) \leq \text{cond}(A).$$

3. Application of the Method:

Consider the following sytem ($x_{exact} = [1, 1, 1]^T$ btw)

$$(5) \quad \underbrace{\begin{bmatrix} 1 & 1 & 0 \\ 0 & 100 & 1 \\ 1 & 0 & 10{,}000 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 2 \\ 101 \\ 10{,}001 \end{bmatrix}}_{b}$$

The coefficient matrix A is almost singular (i.e. $\det(A) \to 0$). The condition number of A, we calculate below:

$$(6) \quad \text{cond}_2(A) = \|A\|_2 \cdot \|A^{-1}\|_2$$

We're considering 2-norm (spectral radius norm) as usual. For this we need to follow the steps below:

1). Compute the largest singular value $\sigma_{max}(A)$
2). Compute the smallest singular value $\sigma_{min}(A)$
3). $\text{cond}(A) = \dfrac{\sigma_{max}(A)}{\sigma_{min}(A)}$

The singular values of a matrix A are the square roots of the eigenvalues of $A^T A$. After I computed

$$(7) \quad \sigma(A^T A) \approx \{1;\ 10{,}000;\ 100000001\}$$

So the singular values are

$$(8) \quad \begin{cases} \sigma_1 = \sqrt{\lambda_1} \approx 1 \\ \sigma_1 = \sqrt{\lambda_2} \approx \sqrt{10,000} = 100 \\ \sigma_3 = \sqrt{\lambda_3} = \sqrt{100000001} = 10,000 \end{cases}$$

So the condition number of A

$$(9) \quad \text{cond}_2(A) = \frac{\sigma_{max}(A)}{\sigma_{min}(A)} = \frac{\sigma_3}{\sigma_1} \approx \frac{10,000}{1} = 10^4.$$

$\text{cond}_2(A) = 10^4$, high condition number indicates that the matrix A is ill-conditioned, meaning that iterative methods might not work well directly on sytem (5). This problem offers the use of multi-scale method.

To use multiscale method, let's transform our equation (5) to

$$(10) \quad \tilde{A}\,\tilde{x} = b \qquad \text{where}$$

$$\tilde{A} = \left[\frac{1}{\|\alpha_1\|_2} \cdot \alpha_1 \quad \frac{1}{\|\alpha_2\|_2} \alpha_2 \quad \frac{1}{\|\alpha_3\|_2} \alpha_3 \right]$$

where α_j, $j = 1, 2, 3$, is the jth column of A in (5).

So

$$(11)\quad \begin{cases} \|\alpha_1\|_2 = \|A[:,0]\|_2 = \sqrt{1^2+0^2+1^2} = \sqrt{2} \\ \|\alpha_2\|_2 = \|A[:,1]\|_2 = \sqrt{1^2+100^2+0^2} = \sqrt{10{,}001} \\ \|\alpha_3\|_2 = \|A[:,2]\|_2 = \sqrt{0^2+1^2+(10{,}000)^2} = \sqrt{100000001} \end{cases}$$

And

$$\tilde{A} = \left[\frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\end{bmatrix} \quad \frac{1}{\sqrt{10{,}001}}\begin{bmatrix}1\\100\\0\end{bmatrix} \quad \frac{1}{\sqrt{100000001}}\cdot\begin{bmatrix}0\\1\\10{,}000\end{bmatrix} \right] =$$

$$(12)\quad = \underbrace{\begin{bmatrix}1 & 1 & 0\\0 & 100 & 1\\1 & 0 & 10{,}000\end{bmatrix}}_{A} \cdot \underbrace{\begin{bmatrix}\frac{1}{\sqrt{2}} & 0 & 0\\0 & \frac{1}{\sqrt{10{,}001}} & 0\\0 & 0 & \frac{1}{\sqrt{100000001}}\end{bmatrix}}_{D} = A \cdot D .$$

for later reference A is the original coefficient matrix A, & D is the diagonal matrix of 2-norms of the columns of A.

In the next steps, we solve the new obtained system in (10) using an iterative method, in our project we focuse on Gauss-Seidel, to find $\tilde{x}$. Once we find $\tilde{x}$, we recover x by scaling back by D:

$$(13)\quad x = D\tilde{x} \qquad \text{by (3).}$$

Let me go through these steps now by expanding (5) $\tilde{A}\tilde{x} = b$ as follows:

$$(14) \quad \left[\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \frac{1}{\sqrt{10{,}001}} \begin{bmatrix} 1 \\ 100 \\ 0 \end{bmatrix} \quad \frac{1}{\sqrt{100000001}} \begin{bmatrix} 0 \\ 1 \\ 10000 \end{bmatrix} \right] \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 101 \\ 10{,}001 \end{bmatrix}$$

By Gauss-Seidel iteration formula in a matrix form

$$(15) \quad x^{(k+1)} = \left[-(D+L)^{-1} \cdot U \right] x^{(k)} + (D+L)^{-1} b .$$

In (15), if we let the starting iteration point $x^{(0)} = [0,0,0]^T$, and using the software to solve, the results in 5 steps with the tolerance $\varepsilon = 10^{-10}$ are as follows:

$$(16) \quad \tilde{x}^{(5)} = \begin{bmatrix} 0.7071\ldots \\ 0.0099\ldots \\ 9.9999\ldots \end{bmatrix}$$

So applying (13) to recover x:

$$(17) \quad x^{(5)} = D\tilde{x}^{(5)} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{10000}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{100000001}} \end{bmatrix} \cdot \begin{bmatrix} 0.7071\ldots \\ 0.0099\ldots \\ 9.9999\ldots \end{bmatrix} \approx \begin{bmatrix} 0.999\ldots \\ 1.0 \\ 1.0 \end{bmatrix}$$

And the relative error for the multi-scale method in <u>5 steps</u> is

$$(18) \quad e_{ms} = \| x^{(5)} - x_{exact} \|_2 = \left\| \begin{bmatrix} 0.999 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\|_2 = 9.769963 \cdot 10^{-15}$$

To verify the reasoning behind the multiscale method, we can compare the condition numbers of A & $\tilde{A}$.

(19) $$\underbrace{\mathrm{cond}_2(A) = 10^4}_{\text{(calculated earlier)}} > \underbrace{2.4143808 = \mathrm{cond}_2(\tilde{A})}_{\text{(calculated in a similar way, software code is in appendix)}}$$

that is matching our formula in (4):

(4) $$\mathrm{cond}(A) \geq \mathrm{cond}(\tilde{A})$$

4. Remarks for the Multi-scale method (specifically for (4) above):

(20) $$\tilde{A} = \begin{bmatrix} \frac{a_{11}}{\|\alpha_1\|_2} & \cdots & \frac{a_{1n}}{\|\alpha_n\|_2} \\ \vdots & \ddots & \vdots \\ \frac{a_{n1}}{\|\alpha_1\|_2} & \cdots & \frac{a_{nn}}{\|\alpha_n\|_2} \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} \frac{1}{\|\alpha_1\|_2} & 0 & \cdots & 0 \\ 0 & \frac{1}{\|\alpha_2\|_2} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\|\alpha_n\|_2} \end{bmatrix} = AD$$

Thus, for simplicity, if we let the condition number to be denoted by 1-norm:

(21) $$d_{max} = \max\{\|\alpha_1\|_2, \|\alpha_2\|_2, \ldots, \|\alpha_n\|_2\}$$

(22) $$d_{min} = \min\{\|\alpha_1\|_2, \|\alpha_2\|_2, \ldots, \|\alpha_n\|_2\}$$

the following holds

property $\|AB\| \leq \|A\| \cdot \|B\|$

$$\mathrm{cond}_1(\tilde{A}) = \|\tilde{A}\|_1 \cdot \|\tilde{A}^{-1}\|_1 = \|AD\|_1 \cdot \|(AD)^{-1}\|_1 = \|A \cdot D\|_1 \cdot \|D^{-1} \cdot A^{-1}\|_1 \leq \|A\|_1 \|D\|_1 \cdot \|D^{-1}\|_1 \cdot \|A^{-1}\|_1$$

$$= \|A\|_1 \cdot \frac{d_{max}}{d_{min}} \cdot \|A^{-1}\|_1 = \frac{d_{max}}{d_{min}} \|A\|_1 \cdot \|A^{-1}\|_1 = \frac{d_{max}}{d_{min}} \cdot \mathrm{cond}(A);$$

That is

$$(23) \qquad \mathrm{cond}(\tilde{A}) \leq \frac{d_{max}}{d_{min}} \mathrm{cond}(A),$$

therefore the condition number of $\tilde{A}$, $\mathrm{cond}(\tilde{A})$ could be no more than that of A, $\mathrm{cond}(A)$.

5. Conclusion:

This project discussed to solve system of linear equations (1) which is ill-conditioned. By carefully scaling the matrix and applying iterative methods, we can achieve better numerical stability and accuracy.

6. Comparisons & appendix:

To compare the results as tables obtained by different iterative methods (namely Gauss-Siedel, Jacobi), and our defined multiscale method, I'll provide a software code for all methods and calculations used in the appendix below as a further reference.